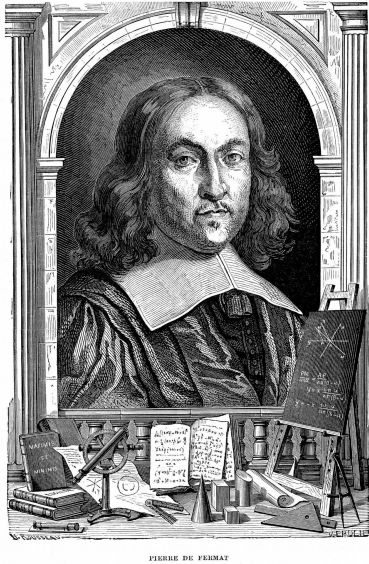


In Claude we Trust – Probability in the 17th and 18th Centuries

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Round 1: Pascal and Fermat v. Descartes

What is the basis of knowledge?



Problem of points.

Q: How to divide the stakes if a game of chance is ended early?

Pascal and Fermat's answer:

Not proportional to the rounds won so far. Proportional to all *possible* ways each could win.

Truth about the world as it is —> truth about the world as it might be.

What is justice?

-Gambling : What is a fair game?

If divided: Change of error

-Interest/Usury: What is a fair rate?

12:0 1/8192

-Juries: What is the best structure?

9:3. About 1/22

8:4. About 1/8

7:5. 2/7

5:3. About 1/4

9:0. 1/1024 (Laplace's Preference)

112:100. About 1/5

501:500 about 1/2

Round 2: Pascal v. Laplace –The Problem of Infinity



St. Petersburg Paradox (Nicolaus Bernoulli - 1713)



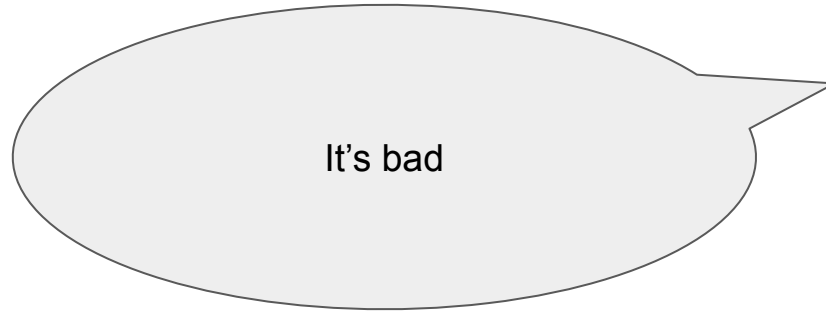
Heads on flip:	You win:
1	\$2
2	\$4
3	\$8
...	...

How much would you be willing to pay to play this game?

Expected Value:

$$E = \left(\frac{1}{2}\right)2 + \left(\frac{1}{2}\right)^2 2^2 + \left(\frac{1}{2}\right)^3 2^3 + \dots$$

Pascal's Wager



Laplace

(Laplace's version: probability of God's existence * gain for belief)

“Probability makes a calculus of common sense.”

-Laplace



Round 3: Daniel Bernoulli v. D'Alembert

The problem of Procrity.



Smallpox

Odds of dying from Smallpox: 1 in 7

Odds of dying from the vaccine: 1 in 100

Q: What is the rational thing to do?

Traditions

Descartes: Rationalism

Pascal: importance of experience

Laplace/Bernoulli: always wager on expected value

LLMs: ????